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An emoji centric approach to sarcasm detection in online discourse

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Sarcasm detection has gained significance in sentiment analysis, especially when social media is rife with cyberbullying and trolling. Emojis have garnered researchers' interest as they are polysemic. But they are usually employed in combination with other modalities in sarcasm detection. This work proposes an emoji-focused approach to study their role in sarcasm detection. This structured approach studies how emojis, sentiment of text and emojis, most frequent emoji in text, and its position helps with sarcasm classification task. Experiments include various machine learning classifiers and BERT fine-tuned for emojis to reveal the decisive role of emojis in sarcasm detection, even in the absence of any other modality. Novel sarcasm-aware GloVe-based emoji embeddings are presented that outperform other available emoji embeddings to achieve highest F1, MCC, and RoC_AuC scores on two unseen datasets. Emoji embeddings, BERT fine-tuned with emojis, and emoji-focused models presented in this work can be used by researchers as baseline for sarcasm classification when situational or conversational context and other modalities like visuals, audio etc. are absent. This emoji-focused approach can be useful in identifying bullying or hateful content on public platforms and even on private chat-based platforms where users may more frequently imply sarcasm under the guise of emojis.

Keywords Sentiment analysis, Sarcasm detection, Emojis in sarcasm detection, Polysemous emojis

Identifying sarcasm is one of the most interesting and challenging tasks in sentiment analysis[1]. In dictionaries sarcasm is defined as, "*a way of using words that are the opposite of what you mean in order to be unpleasant to somebody or to make fun of them*"[2]; "*usually intended to mock or insult someone*"[3]; "*a sharp and often satirical or ironic utterance designed to cut or give pain*"[4]. Sarcasm detection has become an element to identify cases of cyber-bullying[5,6] and trolling[7,8], especially in this era of social media.

The research interest in sarcasm detection focuses on extracting lexical and semantic features like PoS, length of positive or negative phrases, usage of punctuation marks, etc.[1]. Apart from the language-based features, researchers have used various modalities to identify sarcasm like audio, images, and videos[9–11]. Some researchers have also used classifiers that are aware of emojis in the given text. But an emoji-focused approach is rare to find.

But, why emojis? It has been observed that sometimes the users post messages (on social media) without context. This could be due to common knowledge or the latest developments. Ongoing elections, sports events, or some historical context may form the basis of the conversation or a post or message. For instance, consider this tweet by Elon Musk on Twitter (now X), back in 2018. This post was in reference to the cryptocurrency scam, "*Wanna buy some bitcoin* 😉😉" with an anime character in a bitcoin-branded dress. This led to Twitter temporarily blocking Musk's account, even though 😉 indicated the sarcastic nature of this tweet[12]. Therefore, in scenarios where modalities like images, hashtags, or other references are not present to establish context, emojis may be influential in identifying sarcasm. We often encounter such situations on social media platforms or in private chat message exchanges.

Many research experiments employing different modalities have achieved competitive results in sarcasm detection tasks. How emoji-focused sarcasm detection performs in absence of other modalities and conversational or situational context, is yet to be experimented with. Therefore, this work employs an emoji-focused approach that utilizes text and emojis along with their sentiments, placement of emojis, and the most frequently occurring emoji in text for sarcasm classification. We set to answer the following research questions with this work:

- Are emojis effective in detecting sarcasm?
- Can the sentiment of text and emojis (computed using sentiment scores) help in sarcasm detection?

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- Is the frequency of an emoji used in a text an important factor in deciding sarcasm?
- Does the position of occurrence of an emoji in the underlying text play a role in sarcasm detection?

The rest of the paper is organized as follows. The "Related work" section discusses the existing work that uses emojis for sarcasm detection. In the "Methodology" section, experiments across various models with different feature combinations, are discussed. The train and test datasets used for this task are also described. This section presents sarcasm-aware GloVe-based emoji embeddings along with fine-tuned BERT for emoji-focused sarcasm detection. "Experimental results and discussion" section reports results from various experiments conducted. Key observations and detailed analysis of the results are presented in also presented in section. The "Significant contributions" section lists the major contribution of our work. The "Conclusion and future work" section summarizes our findings and discusses the future scope of our work.

Related work

In this section, we discuss approaches employed by different researchers to demonstrate the role of emojis in sarcasm detection tasks. Many researchers have conducted experiments that gauge human reaction and response to emojis in context with sarcasm. Some of them are discussed as follows.

Event-Related Potentials (ERP) experiments were conducted to compute the neural response to conclude that emojis elicit the same response (from the brain) as words for irony delivered by words[13]. To examine the role of emojis in comprehending threatening and indirect responses in *read-reply* sequences, some trials were conducted[14]. Participants in the trial could interpret negative meanings faster when the text contained emojis than the text without emojis. It was demonstrated that winking emoji enables the perception of sarcastic intent[15]. This work also concluded that 😉 is an indicator of sarcasm. It was shown in experiments[16] that shared knowledge (like age or relationship) has a significant effect on how emojis are used to express ambiguous sarcasm. This work demonstrated that younger adults interpreted sarcasm by deriving cues from age and relationships.

Emojis have also been used in supervised settings for sarcasm classification tasks. DeepMoji model[17] learned emoji representations from millions of occurrences of emojis on Twitter. These representations are fine-tuned layer-wise to be further used for the sentiment, emotion, and sarcasm classification. In another work[18] emojis were replaced with their text descriptions to improve irony detection and sentiment analysis tasks. This work reported a 2% improvement in F1-scores over the previous best[19] in irony detection on SemEval Irony Dataset[20] on the corpus with 2396 ironic and 604 non-ironic tweets. A description-based classification may not be appropriate if emoji usage differs across individuals. This is because emoji semantics may change over time or may be used in different contexts[21]. Moreover, in this work, it is not clear how many tweets contained emojis and the average number of emojis per tweet. This makes it difficult to quantify how significant the contribution of emojis is in irony classification tasks. In another experiment GloVe-based word[22] and emoji embeddings[23] are concatenated to demonstrate that emojis can improve sarcasm detection[24]. In a sentence, word embeddings are averaged for all words in a sentence and emoji embeddings are averaged for all emojis in the sentence. These embeddings are then concatenated and then passed through a bi-directional GRU. The resultant vectors are further multiplied with attention weights. They demonstrate significant improvement in F1 scores for sarcasm classification when text and emojis are combined. The emoji embeddings used are based on emoji descriptions. The dataset has 6592 sarcastic and 10,267 non-sarcastic tweets in the Twitter set and 2668 sarcastic and 2803 non-sarcastic tweets in the Facebook set. The researchers use tenfold cross-validation with a 6:2:2 (train-test-validation) split for their experiments. The Facebook set is less imbalanced. Higher F1 scores are achieved on the Facebook set as compared to the Twitter set. It is yet to be seen how this approach performs on a completely unseen data set. It can be further explored whether all emojis contribute towards sarcasm and whether emoji-description-based embeddings can capture the intended use of emojis. SEEmoji MUStARD[25] an extension to multi-modal MUStARD dataset[26] was prepared by manually annotating utterances in MUStARD. These annotations included an emoji, the emoji's sentiment, and the emoji's emotion. A Gated-Emoji-Aware Attention mechanism was used in a multimodal conversation scenario. Like the other experiments, this work too, establishes the importance of emojis in sarcasm detection., Apart from emojis, there are other modalities like video (or image for facial expression), audio (or tone), and conversational context to set up the presence of sarcasm.

Therefore, in this work, along with text, only emojis are used to train the classifiers. The details are presented in the next section.

Methodology

To address the research questions posed we use the following strategy.

Data collection and pre-processing

To train our classifiers we utilize SarcOji dataset[27] which is a sarcasm dataset compiled from five publicly available benchmark datasets for sarcasm. These datasets are Sarcasm Dataset[28], ESD Dataset[24], iSarcasm dataset[29], SemEval2020[30], and SemEval2022[31]. There are a total of 74,276 records in the five benchmark datasets, of which, 31,404 were sarcastic and 42,872 were not sarcastic. In the SarcOji dataset all the text records having emojis were retained. The SarcOji dataset has 29,377 text records with emojis of which 11,448 are sarcastic.

Next, we provide an overview of computation of sentiment scores as detailed in the research paper[27].There are various derived features like sentiment score of text and emojis. To compute text sentiment popular sentiment tools were employed. These tools are VADER[32], TextBlob[33], SentiWordNet[27,34,35]. To compute sentiment score of emojis, the emoji descriptions are used to replace all the emojis in text. The aforementioned tools are then applied to compute emoji sentiment. In case of repeated occurrence of emojis, the intensifier (!) is appended to the emoji text description. For example, if 😭 appears three times in text, the emoji would be replaced by 'loudly crying face!!.' Alternately, the emoji description is repeated and concatenated to capture repeated use

Dataset	Source/compilation method	Sarcastic records	Not-sarcastic records	Total
SarcojiTest1	SPIRS dataset[66]–X posts with the phrase being sarcastic (October, November 2019) Random English posts were scraped from X in the same time period to collect non-sarcastic records	15,000	15,000	30,000
	Posts from X[38] randomly scraped for non-sarcastic and using #sarcasm hashtag for sarcastic posts	50,000	50,000	100,000
	Posts scraped from X[39] using #sarcastic or #sarcasm for sarcastic instances and randomly scraped posts for non-sarcastic instances	35,000	140,000	175,000
SarcojiTest2	Posts scraped from X Sarcastic texts were collected using hashtags # #whatever, #not, #sarcasm, #sarcastic, #wtf, #sarcasmic, #lmao, #sarcasticmemes Non-sarcastic texts were collected randomly	7710	20,000	27,710

Table 1. Details of the number of records collected for SarcOjiTest1 and SarcOjiTest2.

Dataset	Sources	Sarcastic texts (%)	Non-sarcastic texts (%)	Total texts in dataset
SarcOji	Compiled by[27]	39	61	29,377
SarcOjiTest1	Compiled from[37,38], and[39]	52.6	47.4	7228
SarcOjiTest2	Tweets scraped from Twitter (now X)	25	75	5056

Table 2. Distribution of sarcastic and non-sarcastic texts in training and test datasets.

(loudly crying face loudly crying face loudly crying face). Then the sentiment score using VADER, TextBlob, and SentiWordNet are computed for both representations and the higher score is taken as the final sentiment score of the emojis. These three tools are also used to compute the sentiment score of MaxEmoji (the most used emoji in text). The Emoji Sentiment Ranking[36] tool is also employed to compute the sentiment score of the MaxEmoji. A list of emojis used in the text, the most frequently occurring emoji in the text, its position (left, right, or middle) in the text, and its frequency of occurrence in the text are also provided as derived features.

Since, SarcOji dataset is curated from five benchmark datasets, to understand the generalization capability of the proposed models, it was essential to test their performance on unseen data. Therefore, two new test sets were curated to test the performance of the proposed models on the data in the wild.

SarcojiTest1 was compiled using three different benchmark datasets from[37–39]. SarcojiTest2 was compiled by scraping random tweets from Twitter from the year 2021 onwards. The sarcastic tweets were scraped using hashtags like #whatever, #not, #sarcasm, #sarcastic, #wtf, #sarcasmic, #lmao, #sarcasticmemes. For non-sarcastic texts the tweets were scraped without these hashtags. Using sarcasm hashtags allowed quick collection of data as the records were annotated by the authors themselves. But this approach has a caveat too. Not all users may use hashtag based annotation to indicate sarcasm. This may have lead to some sarcastic records landing up in the non-sarcastic pool of records. On the upside the dataset SarcOjiTest2 is a collection of texts as they appear naturally in real-life scenarios. Some users may use hashtags to signal sarcasm while others may not. Hence, testing on this dataset will help understand the proposed model's generalization capability.

Table 1 lists the details of the records collected using the above mentioned approach.

After collecting the data, we processed the data similar to the approach followed for the SarcOji dataset[27]. Only the texts containing the emojis were retained. Features like sentiment scores for text and emojis, the most occurring emoji, and its frequency and position in the text were derived as per the approach given in[27].

Table 2 describes both the training and the test sets.

The datasets were cleaned to remove stopwords, hyperlinks, and characters like newlines, quotation marks, whitespaces, etc. The texts were also converted to lowercase.

Here is an overview of processing steps carried out:

1. *Standardizing labels* Column names were updated to Text and Sarcastic. Sarcasm was labeled as 1 (sarcastic) and 0 (non-sarcastic), standardizing varied original labels like "Yes", "Sarcasm", etc.
2. *Extracting English records* Only English-language records were retained using the Googletrans API. Non-English records (e.g., containing "Bonjour") were removed, though this method may misclassify loanwords (e.g., *café*, *avatar*) commonly used in English.
3. *Transliteration* Transliterated text (e.g., "punch" for पंच or "meet" as a name) may be inaccurately flagged as English by translation tools and thus retained in the dataset.
4. *Removing emoji-less records* Records without emojis were excluded using the Python emoji library and advertools package.
5. *Text cleaning* Mentions, URLs, and hashtags were removed to clean the data.

The processed datasets now contain texts with emojis along with other derived features as in SarcOji dataset. The test datasets are available at https://github.com/VanditaGroverKapila/SarcOjiTestSets .

Feature combinations

The train and test datasets consist of derived features. MaxEmoji is the most frequent emoji in the text. If there is only one emoji in the text, then it is stored as the MaxEmoji. MaxEmojiPos represents whether emoji occurs to

Feature name	F-score
TextSWN	19.97
TextVader	**80.64**
TextTextBlob	38.16
EmojiSWN	399.5
EmojiVader	670.6
EmojiTextBlob	408.44
MEmojiWN	287.69
MEVader	**557.68**
METB	242.38
ESR	**1310.25**

Table 3. ANOVA F-value scores between label and sentiment scores of the text and emojis. Significant values are in bold.

Model number	Model name	Model description—feature combination used
Model 1	PlainText	Text with all emojis removed
Model 2	TextWithEmojis	Text with emojis
Model 3	TextWithEmojisVader	Text with emojis combined with Text and Emoji Vader scores
Model 4	TextMoji	Text without emojis, with MaxEmoji appended at the end of the text
Model 5	TextMojiVader	Model 4 combined with TextVader and MEVader scores
Model 6	TextMojiVaderESR	Model 4 combined with TextVader and ESR scores
Model 7	TextMojiVaderESRPos	Model 6 with MaxEmojiPos
Model 8	TextMojiVaderESREfreq	Model 6 with MaxEmojiNumOccurrence
Model 9	TextMojiVaderESRPosEfreq	Model 6 with MaxEmojiPos and MaxEmojiNumOccurrence
Model 10	PlainTextMojiPos	Text without emojis with MaxEmoji inserted at MaxEmojiPos*
Model 11	PlainTextMojiPosVaderESR	Model 10 with TextVader and ESR scores

Table 4. Descriptions of different models experimented.

the left, middle, or towards the end of the text. MaxEmojiNumOccurrence is the count of the number of times that emoji occurs in the text. The sentiment scores for text, like TextBlob, TextVader, TextSWN, and for emojis, EmojiSWN, EmojiVader, EmojiTextBlob sentiment scores are also stored. MEmojiWN, MEVader, METB, and ESR are sentiment scores computed for emojis.

Since sentiment score features are either derived from text or emojis (except MaxEmojiPos and MaxEmojiNumOccurrence), we used Python's https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.SelectKBest.html[40] to select sentiment features. SelecKBest reports ANOVA's F-value scores between a label (here Sarcastic or Not-Sarcastic) and a feature (here sentiment scores of the text and emojis). We report the scores in Table 3.

We observe from Table 3 that sentiment scores for text and emojis have the highest F-score for TextVader, EmojiVader, MEVader, and ESR. Thus, to experiment we use TextVader for text, EmojiVader (for all emojis), and MEVader and ESR for MaxEmoji.

We experiment with different models using different feature combinations which are described in the following table (Table 4).

MaxEmojiPos refers to the position where the emoji is placed in the plain text. If MaxEmojiPos is 0 then emoji is inserted at the start of the text. If MaxEmojiPos is 1 then the emoji is inserted after the middle word in the text. If MaxEmojiPos is 2 then the emoji is appended at the end of the text.

We refer to models with Text or Text with emojis as the model with NLP inputs. The models having both text and numerical features like sentiment scores, frequency of emoji, or position of occurrence are referred to as models with NLP and Numerical inputs (Table 5).

Word and emoji embeddings

To train our classifiers we experimented with both Word2Vec[41] and *Glove* embeddings[22] for text. For emojis, we used Word2Vec based emoji embeddings *emoji2vec*[23] trained over emoji descriptions. *Emojional*[42] are also Word2Vec based emoji embeddings, but they capture the emotional content of emojis. For Glove-based representations of emojis, we use emoji embeddings available at https://github.com/MirunaPislar/emoji2vec which are based only on emoji descriptions[43]. We call them Glove emoji description embeddings for reference. We also train Glove-based *Sarcasm-aware emoji embeddings* trained on the three datasets SarcOji, SarcOjiTest1, and SarcOjiTest2 datasets. This enabled us to capture emoji embeddings for all the emojis in the datasets. The process for training sarcasm-aware GloVe emoji embeddings is given in Fig. 1.

Model	Embedding combination																							
	Word2Vec [42] + Emoji description (emoji2vec) [22]						Word2Vec [42] + Emojional [43]						GloVe [22] + Emoji description [44]						GloVe [22] + proposed sarcasmaware					
	SarcOjiTest1			SarcOjiTest2			SarcOjiTest1			SarcOjiTest2			SarcOjiTest1			SarcOjiTest2			SarcOjiTest1			SarcOjiTest2		
	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC	F1	MCC	RoC AuC
Plaintext	22.41	0.05	51.98	18.67	0.01	44.23	22.41	0.05	51.98	18.67	0.01	44.23	9.65	0.01	50.39	14.07	0.05	46.83	9.65	0.01	50.39	14.07	0.05	46.83
TextWith emojis	30.31	0.12	55.13	28.57	0.07	49.38	29.07	0.08	54.06	35.57	0.09	50.34	12.97	0.04	51.37	24.59	0.10	49.39	19.26	0.09	53.07	32.48	0.15	50.56
TextWith Emojis Vader	31.51	0.10	55.01	33.57	0.07	50.70	34.80	0.12	56.19	43.40	0.12	52.86	21.09	0.07	52.73	30.97	0.08	49.40	27.82	0.09	53.51	41.55	0.13	51.48
TextMoji	29.47	0.11	54.78	27.57	0.06	49.00	35.00	0.11	56.49	41.43	0.11	52.63	37.59	0.13	56.98	54.08	0.15	55.09	45.22	0.23	62.77	57.62	0.19	57.21
TextMojI Vader	30.58	0.10	54.56	32.86	0.08	50.63	35.00	0.11	56.44	42.24	0.13	52.91	37.45	0.12	56.55	54.17	0.15	54.96	44.98	0.23	62.64	57.54	0.19	57.19
TextMojiVader ESR	30.47	0.10	54.75	33.30	0.08	50.83	35.04	0.13	56.27	42.77	0.13	53.01	38.00	0.13	56.82	54.53	0.15	54.89	45.32	0.23	62.79	**57.93**	**0.19**	**57.31**
TextMoji Vader ESRPos	32.00	0.10	55.76	31.51	0.05	51.01	35.02	0.12	58.02	44.31	0.13	53.27	**42.62**	**0.17**	**59.68**	55.45	0.10	53.06	**48.31**	**0.27**	**65.21**	**58.52**	**0.17**	**56.74**
TextMoji Vader ESREfreq	31.12	0.09	54.65	**32.96**	**0.05**	**49.98**	35.61	0.09	56.22	44.03	0.11	53.89	36.90	0.11	55.90	**54.94**	**0.14**	**54.74**	47.84	0.26	64.77	**59.17**	**0.18**	**57.09**
TextMoji Vader ESRPosEfreq	**33.12**	**0.11**	**55.67**	**33.22**	**0.04**	**49.52**	**39.25**	**0.14**	**58.63**	**47.80**	**0.13**	**55.02**	41.30	0.15	58.64	**55.65**	**0.11**	**53.58**	48.07	0.26	64.97	**59.39**	**0.18**	**57.22**
PlainTextMojiPos	28.73	0.10	54.21	27.33	0.05	48.75	28.29	0.08	53.79	34.81	0.09	50.00	28.66	0.05	52.63	40.31	0.08	50.24	29.88	0.03	51.74	46.94	0.09	51.16
PlainTextMoji PosVaderESR	33.03	0.14	56.35	35.13	0.08	51.27	30.00	0.08	54.18	38.25	0.09	50.86	31.01	0.06	52.75	43.19	0.08	50.65	31.41	-0.01	49.01	46.65	0.01	47.91

Table 5. Comparative performance of all ML based models for different embedding combinations. Significant values are in bold.

The *Sarcasm-aware emoji embeddings* are available at https://github.com/VanditaGroverKapila/SarcasmAwareEmojiEmbeddings.

Classifiers

Each model is trained on different machine-learning algorithms, ensemble classifiers, deep-learning architectures, and transformers The details are as given in follows:

- Machine learning classifiers: Gaussian Naive Bayes, Logistic Regression, Support Vector Machine (SVM)with linear kernel, SVM with polynomial kernel, SVM with RBF kernel, and decision trees
- Ensemble classifiers: Random Forest, AdaBoost, Gradient Boost, and XG Boost
- Deep-learning architectures: For all the experiments we use:
  - Vanilla sequential architecture with a linear stack of layers, LSTM, and Bi-LSTM
  - The hyperparameters for all the deep learning architectures are as follows:
    - Batch size 32
    - Binary cross-entropy loss
    - Sigmoid activation at output layer
    - 25 epochs with early stopping for 6 epochs
  - Models 3 and Models 5–10 have both NLP and numerical inputs (sentiment scores, emoji position, emoji frequency). For NLP we use deep learning architectures as described above and concatenate the output with the output of a vanilla sequential for numerical inputs. The concatenated output is then passed through a dense layer followed by sigmoid for final classification.
- Transformer models like BERT and Tabular BERT have gained popularity in NLP tasks. We use the following transformer-based classifiers for sarcasm detection.
  - BERT for Sequence Classification (base-cased) for PlainText model
  - BERT for Sequence Classification, fine-tuned for emoji tokens in SarcOji dataset for the models TextWithEmojisVader (Model 3), TextMoji (Model 4), and TextMojiPos (Model10)
  - Tabular BERT for sequence classification fine-tuned for emoji tokens in SarcOji dataset

Fine-tuning BERT and tabular BERT

While transformer models like XLM-Roberta-Base support emoji tokens, they may not support all emoji tokens present in our dataset. Hence, we fine-tune BERT for Sequence Classification with the SarcOji dataset. The following hyperparameters were used for fine-tuning:

- Optimizer: Adam
- Learning rate: 2e−05
- Batch size: 32
- Sequence length: 128

The details of the experiment and fine-tuned model are available at FineTuned_Bert. This fine-tuned BERT was used for Model 2, Model 4, and Model 10 where there were no numerical features.

We trained Tabular transformers with the https://multimodal-toolkit.readthedocs.io/en/latest/index.html based on Hugging-Face transformers. We fine-tuned multi-modal transformers with SarcOji dataset along with a combination of numerical features as mentioned in the Feature Combinations section, for Model 3, Models 5 to Model 9, and Model 11.

In the next section we report the results of our experiments.

Experimental results and discussion

In this section, we present the results of the models described with the mentioned classifiers.

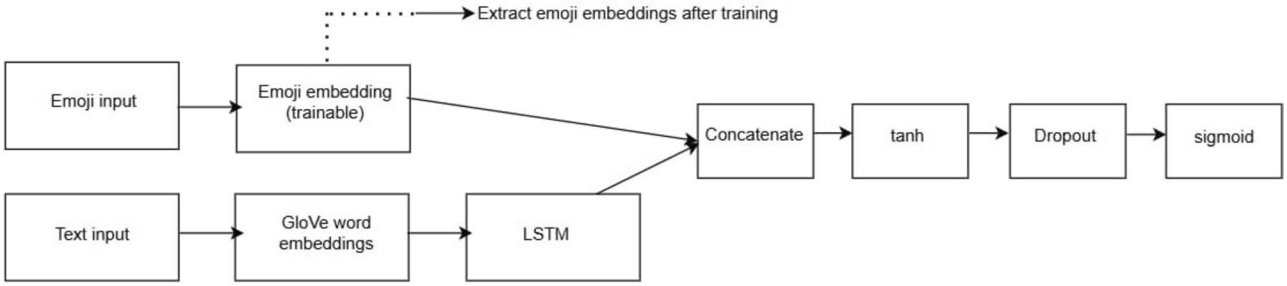


Fig. 1. Training sarcasm aware emoji embeddings.

We report the F1-score[44], Receiver Operating Characteristic Curve (RoC) -Area Under the Curve (AuC) score, and Matthew's Correlation Coefficient (MCC), of all the models experimented with.

$$F1 = \frac{TP}{TP + 0.5 * (FP + FN)} \tag{1}$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)\,(TP + FN)\,(TN + FP)\,(TN + FN)}} \tag{2}$$

F1 and MCC are computed with the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Here we consider Sarcastic Texts as the positive class and Non-Sarcastic Texts are considered as the negative class.

RoC plots the True Positive Rate (TPR) versus False Positive Rate (FPR). AuC of the RoC represents the probability that the model ranks a positive record as positive than a negative record.

F1 is often the metrics of choice in the case of binary classification tasks. Its dependence on the positive class may occlude the classifier performance of the negative class, especially if the dataset is imbalanced.

On the other hand, MCC (ranging from – 1 to 1) uses all the elements (TP, TN, FP, and FN from the confusion matrix. Due to this MCC is now being considered as a more reliable metric[45] as compared to F1. Therefore, F1 and MCC are used to evaluate our classifiers. A higher positive MCC indicates that the classifier is making correct predictions while a negative MCC shows that the classifier is predicting the opposite of true classes. An MCC of 0 means that the classifier is not able to differentiate between the two classes and is making a random guess.

While, the AuC score of 0.5 means the model has a 50% probability of correctly ranking a positive and a negative record. This means the model is randomly guessing the positive and negative labels.

All the 11 models, i.e., the baseline model, Model1 (PlainText), Model2(TextWithEmojis), and the rest of the 9 models with different feature combinations were classified using the ML, ensemble, DL, and transformer-based classifiers. This set of experiments was performed with all the four combinations of embeddings, viz.

- Glove text and GloVe emoji description-based embeddings
- GloVe text and GloVe *sarcasm aware embeddings* (embeddings created in this work)
- Word2Vec text and Word2Vec emoji-description based embeddingsWord2Vec and Word2Vec emojional embeddings

For every model, classifier, and embedding combination, the results reported are *averaged over five runs* for each experiment. The tabulated results are provided in the Supplementary material. Comparative graphs are presented in the following subsection.

ML based classifier performance

The F1-scores, MCC-scores, and RoC_AuC scores of the ten ML and ensemble classifiers were averaged for every model. This was done for all the four embedding combinations for ease of comparison. Since the DL based classifiers had poorer performance, we omit their scores in this discussion.

The following table reports the averaged performance for all embedding combinations across all models.

Discussion on the impact of various emoji-based features on

We now compare the impact of several features that were used to create these models.

Here is a comparison of F1-scores for both SarcOjiTest1 and SarcOjiTest2 for all models and emoji embedding combinations.

Figure 3 reports the comparison of MCC scores.

Take a look at the following figure (Fig. 4) that compares RoC_AuC scores.

Based on these comparative graphs we report our observations.

1. *Impact of including emojis and different text and emoji embedding combinations* We observe that the addition of emojis to text helps in improving the sarcasm detection of the underlying text, on unseen datasets SarcOjiTest1 and SarcOjiTest2, across all models and classifiers, and all combinations of embeddings, as depicted in Figs. 2, 3, and 4. With the addition of emojis, an improvement in F1, MCC, and RoC_AuC scores is observed. This impact is more pronounced when Emojional and *Sarcasm-aware GloVe embeddings* are used for emojis. The classifiers perform poorer when the description-based emoji embeddings are used. In general *Sarcasm-aware GloVe emoji embeddings* presented in this work, achieved the highest F1, MCC, and RoC_AuC scores across all models as compared to the other three embeddings.

Since, *Sarcasm-aware GloVe emoji embeddings* achieved the highest F1-scores, the impact of sentiment scores, most frequent emojis, sentiment scores, position and frequency of emojis are compared w.r.t. *Sarcasm-aware embeddings*. However, the following discussion is applicable on other embeddings as well (ref: Supplementary material).

2. *Impact of sentiment score* A significant improvement in the F1-score was observed on the addition of sentiment scores to text with emojis in Model 3. The improvement in F1-score was more pronounced for SarcOjiTest1.

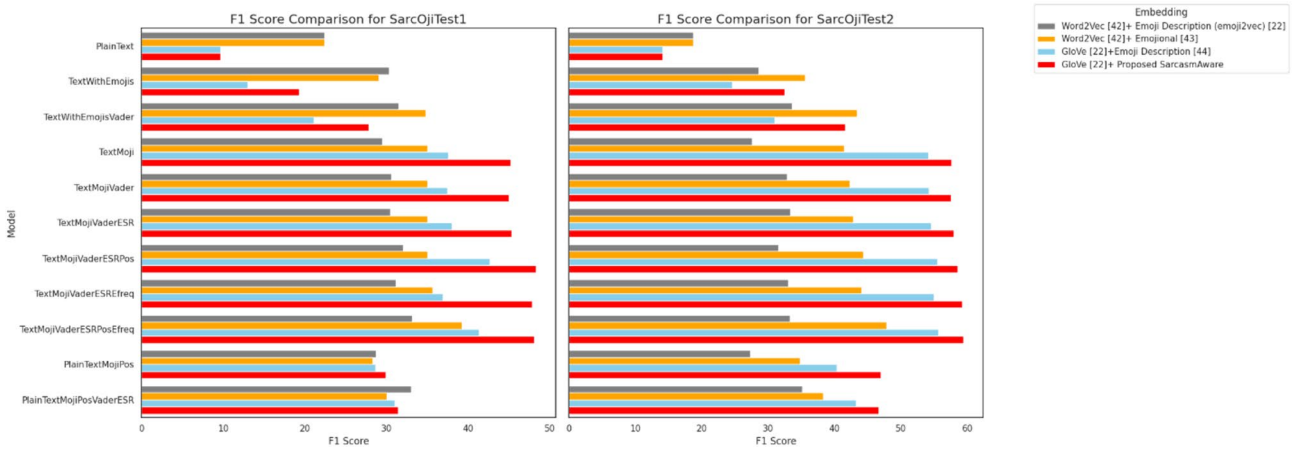


Fig. 2. F1 scores comparison for all models for different embedding combinations.

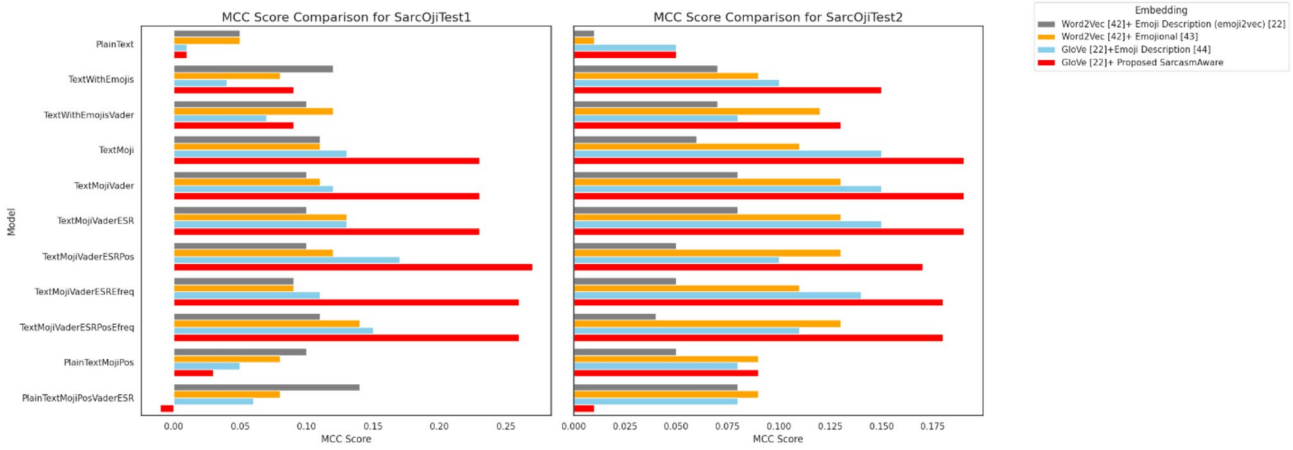


Fig. 3. MCC scores comparison for all models for different embedding combinations.

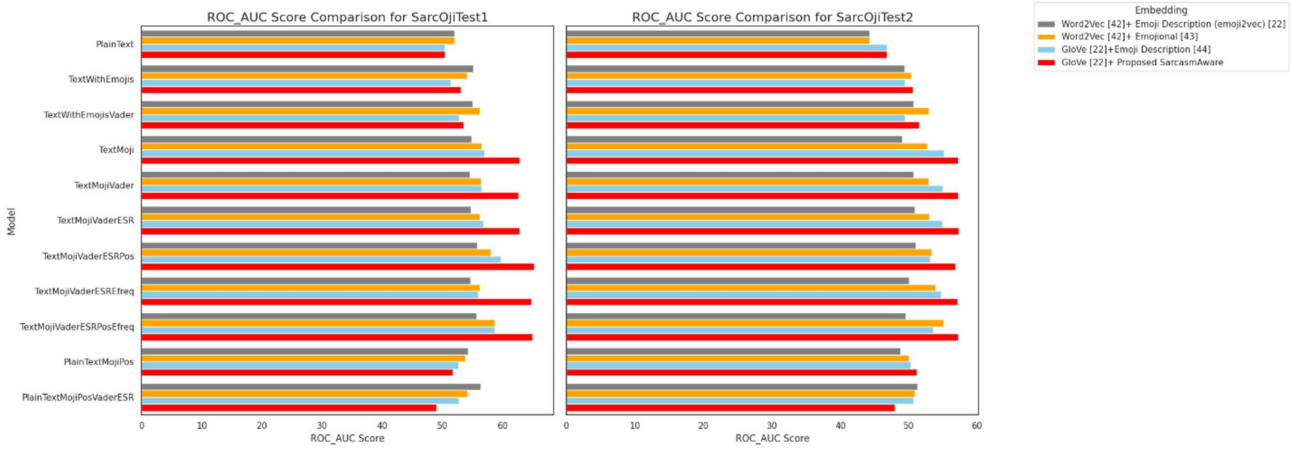


Fig. 4. RoC_AuC scores comparison for all models for different embedding combinations.

3. *Impact of most frequent emoji* Adding the most frequent emoji (or MaxEmoji) at the end of the text (Model 4) improved sarcasm detection significantly. In ML-based classifiers, appending the MaxEmoji at the end of the text (after removing all other emojis) shows an average of 17.4% improvement in F1-scores for SarcojiTest1 as compared to Model3 (TextWithEmojisVader). This improvement is 16.07`% for SarcojiTest2. For transformer-based classifiers, we see a 11.74% improvement in F1-score in SarcojiTest1 and 19.94% in SarcOjiTest2.

Inserting the most frequent emoji at left, middle, or right (based on MaxEmojiPos), in the text (Model 10 and Model 11) did improve F1-scores from the base model (Model 1). But, appending MaxEmoji at the end (Model5) was more impactful than Model 10. Overall, the F1-scores declined drastically for Models 10 and Model 11, when compared with Models 4 to 9.

As the most frequent emoji is significantly improving sarcasm classification performance, we probed the datasets to study the most frequent emoji in the text.

Of the 29377 records in SarcOji dataset, there are 20512 (69.82%) records which have only one emoji. In SarcOjiTest1 3869 records (53.53%) out of 7228 records have only one emoji. In SarcOjiTest2 out of 5056 records, 2684 records (53.09) have only one emoji. Since, almost 70% of records in the training set have only one emoji, this could be a reason why MaxEmoji has a significant impact on classification performance.

4. *Impact of VADER and ESR scores for emojis* We also observed that adding sentiment scores like TextVader and EmojiVader in Model 3 (TextWithEmojisVader) helped improve F1 scores when compared to Model1 (PlainText) and Model 3 (TextWithEmojis). across all embeddings. While there is an improvement on addition of ESR, that is in Model6 (TextMojiVaderESR) when compared Model4 (TextMoji). This improvement is minor, i.e., a 0.4% increase in F1 score in SarcOjiTest1 and 1.14% increase in F1 score in SarcOjiTest. The impact of sentiment scores on F1-score in Tabular Transformers based classifiers was negative.
5. *Impact of MaxEmojiPos and MaxEmojiNumOccurrence* A minor improvement in F1 scores was observed on adding MaxEmojiPos and MaxEmojiNumOccurrence from Model 7 to Model 9) in ML and ensemblebased classifiers when compared to Model4(TextMoji). For SarcOjiTest1 the improvement in F1 score is 3.28% (Model7—TextMojiVaderESRPos) and 2.52% (Model8—TextMojiVaderESREfreq). A decline of 0.19% in F1 score is observed in Model9 (TextMojiVaderESRPosEfreq). 1.48% when compared to TextMoji model F1-score. In SarcOjiTest2, a minor improvement of 0.98% (Model7), 1.64% (Model8), and 1.74% (Model9) is observed over Model4. We saw a decline in F1 scores, when we added MaxEmojiPos and MaxEmojiNumOccurrence for the Tabular Transformer classifiers. Although adding numerical features help, in slightly improving sarcasm detection, this effect is not very significant.

We further looked at the position of the MaxEmoji in the sarcastic and not sarcastic records in the training and the test sets. The % of emojis occurring towards the beginning, middle, and the end of the text are reported in Table 6.

It is observed that fewer records (sarcastic or non-sarcastic) have emojis towards the beginning of the text. In general, more records have emojis occurring towards the end. In the training dataset (SarcOji) 18.89% sarcastic records have MaxEmoji in the middle of the text. But more records in the test datasets have MaxEmoji in the middle. The test datasets have different usage patterns when we look at the position of MaxEmoji. This could be a reason that the Models 10 and 11 (for ML and ensemble classifiers) see a dip in F1-scores as the MaxEmoji is inserted at the position (beginning, middle, or end) (Fig. 5).

Now, we look at the frequence of occurrence of MaxEmoji for the three datasets. Table 7, reports % of records where an emoji is used more frequently (more than one occurrence).

In the training dataset the number of sarcastic records with more than one occurrence of MaxEmoji (MaxEmojiNumOccurrence > 1) is double than the non-sarcastic records which have MaxEmojiNumOccurrence > 1. This pattern is not seen in the test datasets. Thus, Model 8 and Model 9 do not see a significant improvement in F1-scores.

Classifier performance—a comparison using F1, MCC, and RoC_AuC scores

The highest F1 score (48.31%) averaged across ten ML and ensemble classifiers for SarcOjiTest1 is obtained for Model7. For SarcOjiTest2, the highest averaged F1 score is 59.39% for Model9. The FineTuned BERT reports the highest F1-score of 56.85% for SarcOjiTest1 and 65.32% for SarcOjiTest2 for Model4.

The F1-scores for ML and ensemble classifiers appear to give a competitive performance in case of SarcOjiTest2, which is the dataset compiled using random posts from X. MCC for the Sarcasm-aware GloVe

	Sarcastic			Not sarcastic		
Dataset	Begin	Middle	End	Begin	Middle	End
SarcOji	13.75	18.89	67.36	6.9	12.76	80.34
SarcOjiTest1	4.18	24.97	70.85	13.71	30.84	55.44
SarcOjiTest2	20.11	41.25	38.64	8.78	31.66	59.56

Table 6. % of records with emojis occurring towards the beginning, middle, and the end of the text in the training and text datasets.

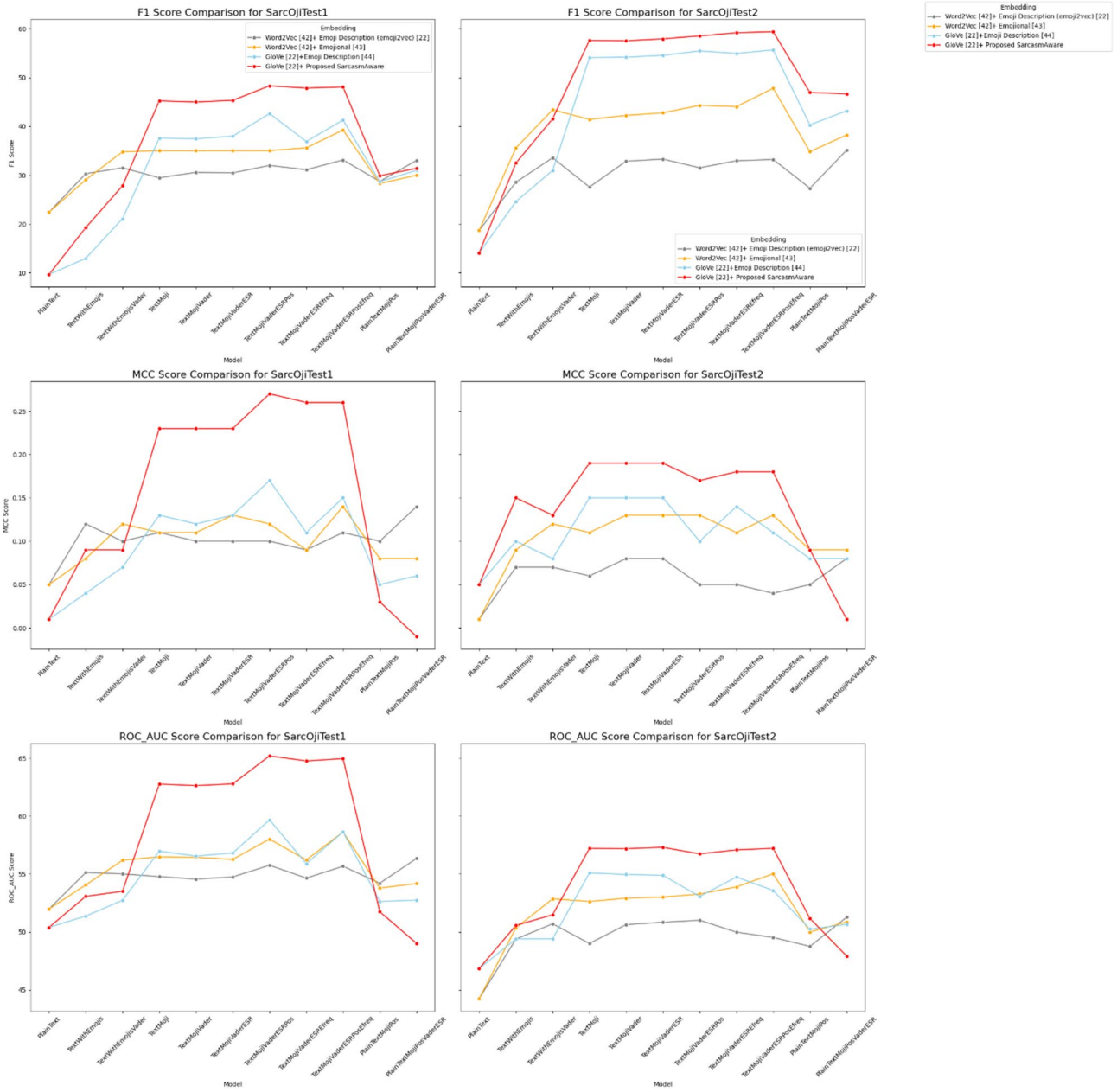


Fig. 5. Comparison of model performance across embedding combinations on SarcOjiTest1 and SarcOjiTest2.

Dataset	Sarcastic	Not sarcastic
SarcOji	23	11.6
SarcOjiTest1	18.97	25.8
SarcOjiTest2	25.97	27.02

Table 7. % of records that have more than one occurrence of MaxEmoji.

emoji embeddings is between 0.23 and 0.27 for models 4 to 9 on SarcOjiTest1 dataset. While for the rest of the embeddings, the MCC ranges between 0.09 and 0.15. For SarcOjiTest2 the MCC is between 0.08 and 0.15 and for the Sarcasm-aware GloVe emoji embeddings it is slightly higher (0.15 to 0.18). A classifier that performs well for all the four categories (TP, TN, FP, FN), will have a higher MCC score[45]. In general, RoC_AuC scores are higher for SarcOjiTest1 (65.21% for Model 7). The reason for higher MCC and RoC_AuC scores is that the models are able to better generalize across both the classes. For both the datasets, the Sarcasm-aware GloVe embeddings higher RoC_AuC scores. The addition of numerical features helps in the improvement for RoC_AuC scores

for ML and ensemble classifiers SarcOjiTest1. In general we were able to observe that emojis in text helped in improving the performance of sarcasm classifiers and addition of numerical features helped improve the performance slightly.

Transformer based classifier performance

Figure 6 shows the comparison of F1, MCC, and RoC_AuC scores.

The transformer-based classifiers have much higher value for both F1, MCC, and RoC_AuC scores. TextMoji model achieves the highest scores. These scores being 56.85, 0.42 and 71.38 for SarcOjiTest1. For SarcOjiTest2, the TextMoji achieves 65.32(F1), 0.39 (MCC), and 69.1 (RoC_AuC) for SarcOjiTest2. This shows that the transformer based classifiers significantly improved sarcasm classification performance when emojis were added to text. In general they have a better generalization capability too across both the datasets.

Overall performance of classifiers

The best F1-score 50.88%, MCC score of 0.31, and RoC_AuC score of 67.41% for SarcojiTest1 is achieved by Logistic Regression trained with GloVe text and sarcasm aware emoji embeddings with Model 7 (TextMojiVaderESRPos model). For SarcojiTest2 Gaussian Naïve Bayes trained with GloVe text and sarcasm aware emoji embeddings gives the highest F1-score of 66.19%, MCC score of 0.25, and RoC_AuC of 62.04% with Model 7 (ref: Supplementary material).

The transformer-based Model 4(TextMojiVaderESREfreq) gives F1-score of 56.85% on SarcOjiTest1, with RoC_AuC of 71.38% and with MCC of 0.42. This model also achieves highest F1 (65.32%), RoC_AuC (69.1%), and MCC (0.39) scores for SarcOjiTest2. In general, Transformer-based models give better MCC scores across all models. Therefore, in future, techniques like data augmentation, fine-tuning hyperparameters etc. can be explored to further improve the performance of transformer-based models.

In case of Fine-tuned Transformer model, Model 10 observes a significant increase (5.04% to 7.73%) when compared to models with numerical features (Models 5, 6, 7, 8, and 9).

In general classifiers' performance from Models 1–9 show that addition of emojis improves sarcasm detection. Addition of numerical features does not significantly improve the performance of the classifiers.

Error analysis

To test the models, we analyze some random text records on Models 5 and 6 as they use different emoji sentiment scores. And Models 7 and 9 as they perform the best on both the datasets. We used Logistic Regression for inferencing because of its simplicity and its comparable performance with ensemble classifiers.

These results are reported in Table 6.

In Table 4, '0' indicates that the text was classified as not sarcastic while a '1' indicates that the text was classified as sarcastic.

Here are some observations from Table 8:

- "I like this movie" was classified as not sarcastic by all the models. On the other hand, "I like this movie ♡♡ " was classified as not-sarcastic by all but one classifier.
- ♡ is classified as sarcastic by all the classifiers.
- Model 6 misclassifies, "I like this movie ♡♡" as sarcastic. One reason could be the occurrence of the word 'like' in the training dataset. Upon analyzing the SarcOji dataset, it was discovered that the word '*like*' had 857 occurrences in 11,448 sarcastic records (7.59%) and 1248 occurrences in 17,929 (6.96%) non-sarcastic records. This co-occurrence of frequent words/tokens in SarcOji dataset, viz. 'like' and ♡ could have contributed to this sarcasm classification.
- "I like this movie 🙄🙄🙄🙄" is misclassified by all the classifiers apart from Model 9 which has MaxEmojiNumOccurrence (the frequency of MaxEmoji) as a feature. The SarcOji dataset has 21.75% of occurrences of 🙄 in sarcastic texts[27], yet the models are unable to capture this information.

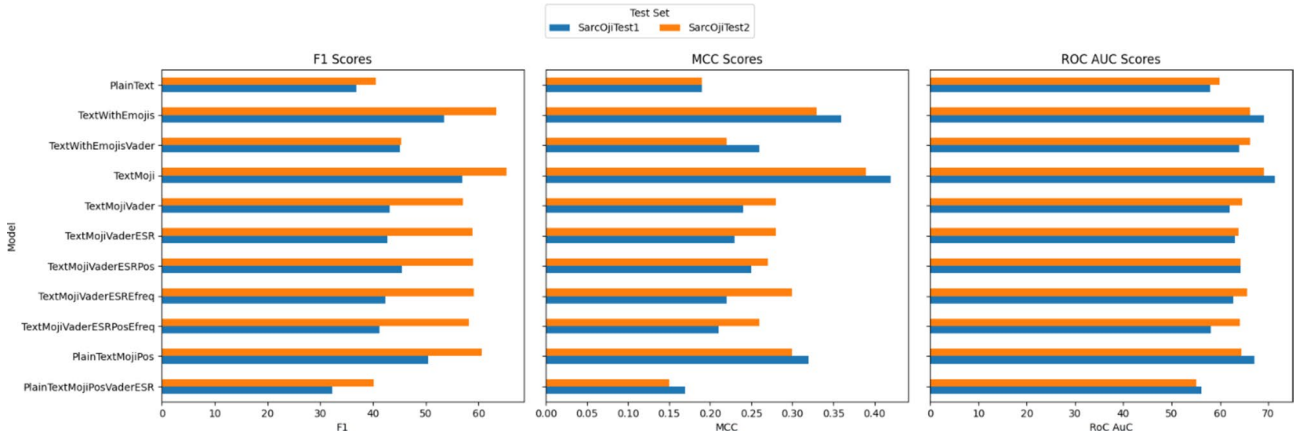


Fig. 6. Performance of transformers based classifiers.

Text	Model 5 TextMojiVader	Model 6 TextMojiVaderESR	Model 7 Model6 + Pos	Model 9 Model7 + MaxEFreq
I like this movie	0	0	0	0
😊 I like this movie	**0**	**0**	1	1
I like this movie ♡♡	0	1	0	0
I like this movie 😍😍😍😍	**0**	**0**	**0**	1
I love this movie	**0**	**1**	**1**	**1**
I love this movie 😍😍	**0**	**1**	**1**	**1**
😢😢😢 What a good use of my time	0	0	1	1
😊	1	1	1	1
😠😠 their latest burger is superb	1	1	1	1
i liked their burger 😠😠	0	0	0	0
it was a bad day 😢	0	0	0	0
it was a bad day 💩	0	0	0	0
lol 💩	1	1	1	1
Lol	1	1	1	1
💩	1	1	1	1
Watching the Olympics	0	0	0	0
Watching the Olympics ♡	0	0	0	0
Watching the Olympics ❤	0	0	0	0
Watching the Olympics ♡♡	0	0	0	0
♡	1	1	1	1
we can meet after	0	0	1	1
we can meet after 😉	**0**	1	1	**0**
we can meet after 😊	**0**	0	0	0
😊 we can meet after	**0**	0	1	1
we can meet after 😍	**0**	1	**0**	**0**

Table 8. Inference of Logistic Regression on random text records for Models 5,6,7,9 Significant values are in bold.

- The records having the word 'love' were classified as sarcastic by three of our models. Love occurs 1126 times (9.84%) in sarcastic records and 1371 (7.65%) in non-sarcastic records. It is possible that the users, use the word 'love,' to convey a sarcastic intent.
- Models 7 and 9 have MaxEmojiPos (position of first occurrence of the MaxEmoji) as one of the features. These models sometimes misclassify non-sarcastic texts, but they can capture sarcastic intent if the emojis occur towards the beginning of the text. This is not seen when emojis are placed towards the end or in case of frequent (emotional) words like 'like' or 'love.'

With these observations, we infer that our models can capture sarcastic intent of the emojis. But if the text corpus has words that occur more frequently (e.g. the words like 'like' or 'love' in this case) they may not necessarily be able to capture sarcastic (or not) intent of the user. One reason for this error could be the use of GloVe embeddings for words. Although, the emoji embeddings are sarcasm aware, the word embeddings used for word representations are not. Rather, the GloVe word embeddings are primarily based on co-occurrence statistics of the words found in the text corpus created from Wikipedia 2014 data and English Gigaword fifth edition[46]. Therefore, it can further be explored by using word embeddings that capture not just the statistical co-occurrence of words but also capture their relationship with surrounding text.

One such model is BERT, that captures the left and right context of a word (token) to create embeddings. This is discussed in the next section.

Fine-tuned BERT for emoji focused sarcasm detection

The experiments presented in this study helped conclude the significance of emojis in detection of sarcasm. BERT fine-tuned on the training (SarcOji) dataset achieved higher MCC scores (for models with text features) as compared to ML models. The tabular-BERT models (models with text and numerical features) too achieved higher MCC scores in comparison with ML models. But a significant improvement in F1 and MCC scores was not seen when the numerical features were added. Hence, in future, it is worth exploring if more training and creating a custom BERT model to capture numerical features can further improve performance of the proposed approach.

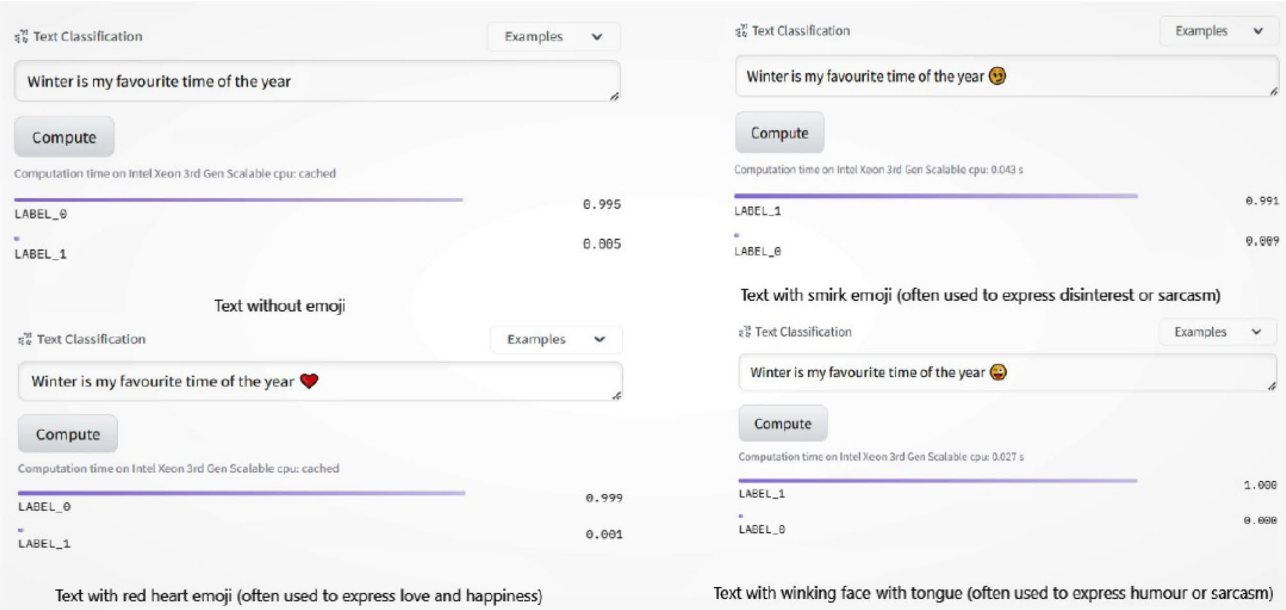


Fig. 7. Sarcasm classification examples using BERT fine-tuned with emojis.

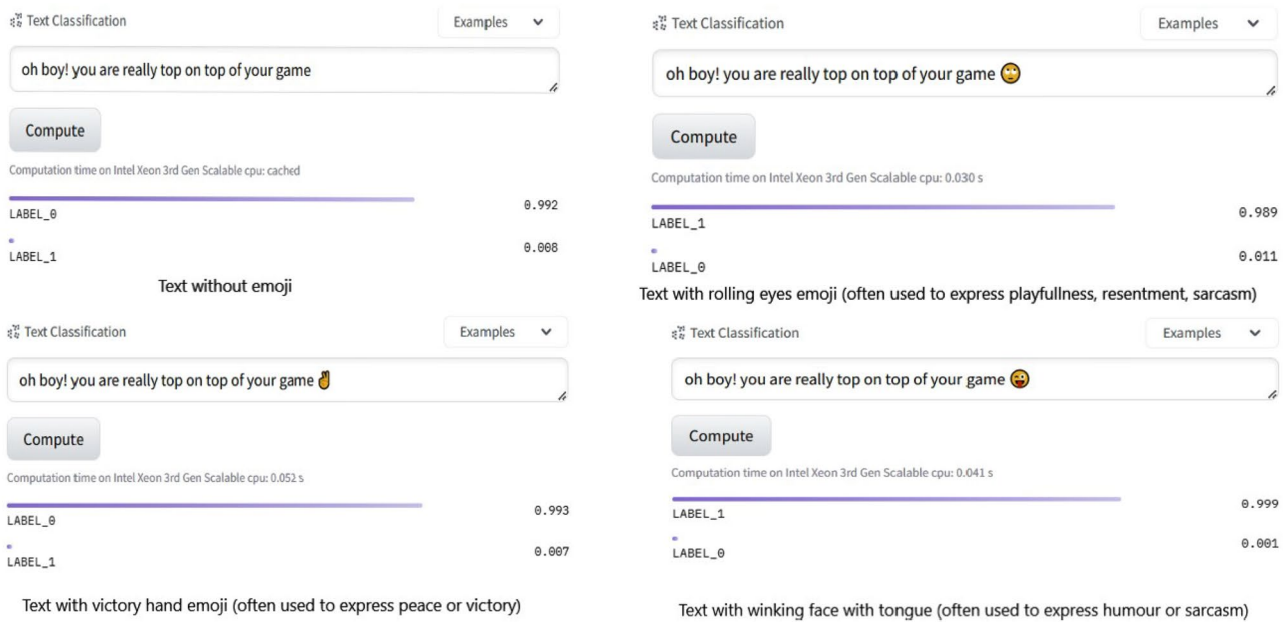


Fig. 8. Sarcasm classification examples using BERT fine-tuned with emojis.

We use the Hugging-Face inference API to demonstrate how emojis enable sarcasm classification with the help of a few examples tested on BERT Fine-tuned with SARC in Figs. 7 and 8.

Label_0 denotes the probability of the text being non-sarcastic and Label_1 denotes the probability of the text being sarcastic.

Since BERT is trained on a larger dataset, fine-tuning it for sarcasm detection using emojis can enable us to capture more nuanced textual context. In future, this fine-tuned BERT model can be customized to train a downstream sarcasm classification task with emojis.

Significant contributions

In this work of emoji-focused approach to sarcasm-detection we experimented with several models to observe the impact of emojis in sarcasm detection, especially in absence of any situational or conversational context and other modalities like images, hashtags, audio/tone, mentions, etc. The models proposed included emojis,

text and emoji sentiment, placement, and frequency of emojis in the text. We were able to address the research questions posed at the beginning of this paper.

1. Addressing the Research Questions on Role of Emojis in Sarcasm Detection: We answer each of the research questions based on the results reported in the previous section.
   - Are emojis impactful in detecting sarcasm?
     - Yes. The inclusion of emojis in the text led to an increase in F1, MCC, and RoC_AuC scores across all models when compared to PlainText (Model1).
   - Can the sentiment of text and emojis (computed using sentiment scores) help in sarcasm detection?
     - Adding sentiment scores of (words in) text and emojis improved F1, MCC, and RoC_AUC scores for all the word and emoji embedding combinations in ML and ensemble based classifiers. The transformer-based models saw a decline in all the metrics on inclusion of sentiment scores using the Multimodal(Tabular) BERT model. was used with the text.
   - Is the frequency of an emoji used in a text an important factor in deciding sarcasm?
     - MaxEmoji or the most-frequent emoji plays a critical role in the sarcasm detection task. MaxEmoji-based models performed better than the models where all emojis in the text were used for classification.
     - One of the reasons for MaxEmoji based models performing better was that users used either a single emoji or in case of the multiple emojis in text.
   - Does the position of occurrence of an emoji in the underlying text play a role in sarcasm detection?
     - Models where MaxEmoji was appended at the end, performed significantly better than the models where MaxEmoji was inserted at the position of the first occurrence (left, right, or middle). One of the reasons for this result could be that 67.36% of records in the training set contained MaxEmoji towards the end.
     - In some cases, the models which captured emoji position and frequency of emoji occurrence, were able to identify the sarcastic intent if the emoji was placed in the front of the text. But these models also misclassified some non-sarcastic texts and had a lower F1 and MCC score.
2. Sarcasm Aware GloVe based Emoji embeddings: Apart from addressing the research questions posed above, we have been able to prove that Sarcasm-aware emoji embeddings play a crucial role in discerning sarcasm. Classifiers trained with Sarcasm-aware GloVe based emoji embeddings along with GloVe text embeddings outperformed all other classifiers. We present these GloVe-based Sarcasm Aware emoji embeddings for future research to the research community (Sarcasm Aware Emoji Embeddings Github).
3. Emoji-focused Test Datasets: We also make available test datasets SarcojiTest1 and SarcojiTest2 to the research community for future experiments. These test sets are available on Test datasets Github.
4. Fine-Tuned Bert for Emoji-focused Sarcasm Detection: We present Bert(base-cased) fine-tuned with emojis for emoji-focused sarcasm detection at Finetuned BERT HuggingFace.

Conclusion and future work

The rapid and exponential increase in online discourse has a potential of significant impact on human behavior. There have been several studies that have noted hostile discussions[47,48], online harassment[49–51], toxicity[52–54], trolling[55–57], and hate speech[58–60]. Often sarcasm under the guise of positive messages is used to communicate hate and bully people in online discourse. Other implications are misinterpretation of events when satirical version of news is interpreted as true leading to proliferation of fake news. In face-to-face communication, facial expressions, tone, and gestures often help people interpret sarcasm. Emojis have a strong potential to provide visual cues to communicate sarcasm in online discourse. The use of emojis for cyberbullying, display of aggression, and express sarcasm and irony have been reported in many studies[61–65]. It may also be noted that emojis have now proliferated workspaces, making their way through email reactions and formal channels of communication like team chat groups on social media platforms like Microsoft Teams, Slack, WhatsApp etc. Hence, a study on using emojis to detect sarcasm is a pioneering step at the right time as use of emojis continues to evolve in online discourse.

This study investigates the impact of emojis in detecting sarcasm in text, independent of other contextual cues. Through experiments with various models, the research confirms that emojis are significant in identifying sarcasm, with their inclusion boosting model performance. The sentiment of both text and emojis generally improved detection, though not for transformer-based models. A key finding is the critical role of the most frequent emoji ("MaxEmoji") in a text; models using only the MaxEmoji outperformed those considering all emojis. Furthermore, the position of the MaxEmoji matters, with placement at the end of the text yielding the best results.

Beyond these findings, the research makes several contributions to the field. It introduces "Sarcasm-aware GloVe based emoji embeddings," which proved highly effective. We also release two new test datasets, SarcojiTest1 and SarcojiTest2, and a fine-tuned BERT model specifically for emoji-focused sarcasm detection.

In the future, we propose to work on further experimenting with Fine-Tuned BERT on emoji focused sarcasm classification task to further improve the classification results. Further research can be conducted to utilize the

proposed models proposed in this work in understanding online communication, where sarcasm can mask hostility and contribute to misinformation.

Data availability

The Sarcasm Aware Emoji Embeddings, training and test datasets for this work are placed at https://github.com/VanditaGroverKapila. The test datasets used for this work are placed at https://github.com/VanditaGroverKapila/SarcOjiTestSets. The BERT fine-tuned with emojis for sarcasm detection is available at https://huggingface.co/Vandita/Bert-finetuned-Sarc

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Author contributions

Vandita Grover contributed to the acquisition, analysis, modeling and experiments, interpretation of results, and drafting of the manuscript. Hema Banati contributed to conception and design of the work, interpretation of results, and review and revision of the manuscript. Both the authors have approved the submitted version and agree to be personally accountable for their contributions and integrity of any part of this work.

Declarations

Competing interests

The authors declare no competing interests.

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